Mini-review Article

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A steady state model of agricultural waste pyrolysis: A mini review

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Abstract

Agricultural waste is one of the main renewable energy resources available, especially in an agricultural country such as Serbia. Pyrolysis has already been considered as an attractive alternative for disposal of agricultural waste, since the technique can convert this special biomass resource into granular charcoal, non-condensable gases and pyrolysis oils, which could furnish profitable energy and chemical products owing to their high calorific value. In this regard, the development of thermochemical processes requires a good understanding of pyrolysis mechanisms. Experimental and some literature data on the pyrolysis characteristics of corn cob and several other agricultural residues under inert atmosphere were structured and analysed in order to obtain conversion behaviour patterns of agricultural residues during pyrolysis within the temperature range from 300 °C to 1000 °C. Based on experimental and literature data analysis, empirical relationships were derived, including relations between the temperature of the process and yields of charcoal, tar and gas (CO_2, CO, H_2 and CH_4). An analytical semi-empirical model was then used as a tool to analyse the general trends of biomass pyrolysis. Although this semi-empirical model needs further refinement before application to all types of biomass, its prediction capability was in good agreement with results obtained by the literature review. The compact representation could be used in other applications, to conveniently extrapolate and interpolate these results to other temperatures and biomass types.

Keywords

Agricultural wastes, corn cob, pyrolysis, products, steady state, modelling

Introduction

Ever increasing energy demand and the climate change problem caused by anthropogenic greenhouse gas emissions have resulted in the worldwide effort to find a sustainable and environmentally friendly alternative to today's fossil fuel-dominated energy supply. There is potential for biomass to be useful in solving some of the world's energy and environmental problems, as it is widely recognised as an environmentally friendly and renewable energy source. Biomass is a promising energy source owing to its abundance, carbon-fixing and carbon-neutral properties (Chen and Kuo, 2011; Deng et al., 2009; Fiaschi and Carta, 2007; Lu and Chen, 2013; Trninić et al. 2012; Wang et al. 2011; Werther et al., 2000). Biomass is very useful in meeting different kinds of energy needs, including fuelling vehicles, providing process heat for industrial facilities, generating electricity and heating homes (Demirbas, 2009). Examples of such biomass include agricultural residues, forest residues and food processing waste.

Agricultural residues are one of the main renewable energy resources available, especially in an agricultural country such as Serbia. In Serbia, agricultural residues (including abundant by-products from agricultural industries as well as residues from post-harvest such as bagasse, corncob, etc.) are commonly available for use as alternative energy sources. It is estimated that every year in Serbia, a total amount of 4.24 million tonnes of agricultural residues is produced, which is equivalent approximately to 1.71 million tonne of oil equivalent (toe) (Stojiljkovic, 2013; Trninic, 2015). In particular, corn cob as an agricultural residue has statistically been shown to have high resource potential in Serbia (1.5 million tonnes of corn cob per year, or 0.7 million toe (Trninic, 2015)).

Several thermochemical techniques, such as pyrolysis, gasification and combustion processes, have been proposed for biomass conversion into hydrocarbon fuels (Bridgwater and Grassi 1991; Bridgwater, 1995, 1996, 1999, 2001, 2003; Bridgwater and Boocock, 1991; Bridgwater and Cottam, 1992). Among these techniques, pyrolysis has attracted particular interest as it can directly convert solid biomass into a bio-oil, a gaseous fuel and a granular charcoal. The pyrolysis oil (bio-oil) derived from biomass could act as feedstock for producing hydrocarbons that could be readily integrated into existing petroleum refineries or

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future bio-refineries (Kim et al., 2013). The produced gas can be used as a fuel for synthetic natural gas production or used for production of liquid fuels (e.g. methanol, ethanol, gasoline, aviation fuel and diesel fuel). The charcoal can be used as an active carbon, as a reducing agent in the metallurgical industry (e.g. to smelt metal ores), as a fuel in households (e.g. as a barbecue charcoal), as an adsorbent or as a raw material for the production of chemical compounds (carbon–disulphide (CS_2), calcium–carbide (CaC_2), silicon–carbide (SiC), sodium cyanide (N_aCN), as a soil fertiliser, carbon black, various pharmaceutical compounds, etc.) (Antal and Mok, 1990; Antal et al., 1996, 2000; Becidan et al., 2007; Yaman, 2004). Also, pyrolysis is interesting, not only as an independent process leading to the production of energetically dense products, but also as an intermediate step in a gasification or combustion process (Bridgwater, 1993).

Compared with combustion, which is widely practised commercially, and gasification, which is being extensively demonstrated around the world, pyrolysis of biomass is still at the stage of being developed for commercial use, and needs to overcome a number of technical and economic barriers to compete with traditional fossil fuel-based techniques (Jahirul et al., 2012), although numerous projects have been promoted. In spite of the fact that mankind has been manufacturing charcoal for about 6000 years, traditional methods for charcoal production in developing countries realise yields of 20% or less, and modern industrial technology offers yields of only 25%–37% (Antal et al., 1996). From a theoretical perspective, charcoal production should be efficient and quick (Antal et al., 1996). Based on thermochemical equilibrium calculations, the yield of charcoal from most biomass feeds, at atmospheric pressure, should be in the approximate range of 55% (corn cobs have a carbon content of 45%) (Wang et al., 2011). Also, industrial-scale thermochemical production of bio-oils, by fast or flash pyrolysis, is costly and technically complex (Radlein and Quignard, 2013) although it has been adopted for commercial practice (e.g. the Empyro plant in Holland, Avello in the USA, etc.). In this regard, the development of thermochemical processes requires a good understanding of pyrolysis mechanisms. This includes determination of the most significant pyrolysis parameters and their effect on the process, plus determination of the pyrolysis products (complex mixtures of the volatile matter and charcoal).

The literature contains many experimental studies of biomass pyrolysis, regarding mechanisms, kinetics and product distribution. Different experimental set-ups have been used and the operating conditions, fuel type, methodologies and measurements have varied widely among investigations (Neves et al., 2011). Given the complexity of the conversion process, it is difficult to characterise it based solely on the experimental data. Therefore, mathematical modelling of pyrolysis, based on general trends for both product distribution and properties as a function of pyrolysis parameters (e.g. temperature), give a better understanding of the conversion process. Therefore, the objective of a mathematical model of pyrolysis is to predict the behaviour of a pyrolysis process and to work out ways to control its course.

The objectives of this study were to investigate the pyrolysis of corn cob (and other agricultural residues) and to provide a guide for the pyrolysis characteristics of biomass, to determine the yields of charcoal, tar and volatiles produced during corn cob pyrolysis, and to determine the composition of the light gas during conditions typically found in pyrolysis (300 °C–950 °C).

An empirical model was developed to describe the general trends of product distribution as a function of temperature, and which consisted of elemental and mass balance (based on elemental and proximate analyses), energy balance and empirical relationships. The empirical relationships between product yields and temperature were derived from the data from experiments in this study and the literature, which addressed pyrolysis of agricultural residues. This empirical model compiled the collected experimental data in a structured tool that can be effectively used to analyse the biomass pyrolysis process (Neves et al., 2011). The presented model was validated with open circles experimental data from other studies (Cao et al., 2004; Ioannidou et al., 2009). Considering that pyrolysis is a key conversion stage during gasification and combustion, the proposed model can be implemented in biomass gasification and combustion mathematical models.

Experimental study

Samples

Grab samples of corn cob were obtained from Surčin, in the Belgrade municipality in Serbia (ZP Maize Hybrid, ZP 505).

Proximate and ultimate analyses of corn cob were shown in recent works that investigated the charcoal yields from corn cob and kinetics of corn cob pyrolysis (Trninić et al., 2012; Wang et al., 2011). Also, our corn cob feed samples were subjected to proximate analysis according to American Society for Testing and Materials (ASTM) methods E871 and 872 (ASTM 871-82, 2013); however, the ash content of the feeds was determined according to ASTM D1102-84 (ASTM D1102-84, 2013). Elemental analyses of the samples were conducted using an elemental analyser (Vario MACRO Elementar) according to standards ASTM E777-08 (carbon and hydrogen) (ASTM E777-08, 2008), ASTM E778 (nitrogen) (ASTM E778-15, 2015) and ASTM E775 (sulphur) (ASTM E775-15, 2015). The oxygen (O) content was determined by the difference of 100% and the sum of the ash, carbon (C), hydrogen (H), nitrogen (N) and sulphur (S) contents. Table1 summarises the properties of the corn cob used, and shows it could be considered as a typical high volatile, low fixed carbon and low ash biomass fuel.

Thermogravimetric experiments

The corn cob samples were ground in a cutting mill mounted with a 1 mm sieve to produce corn cob powder. Then, corn cob powder samples were left to dry in an oven at 105 °C for 24 h prior to experiments. The use of small-sized particles was to minimise the influence of diffusion and heat transfer problems inside the particles. The omission of the finest particles eliminated the problem of dusts blown out by the gas stream during the experiments.

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Table 1. Proximate and elemental analysis of corn cob.

Elemental analysis[a] (wt.%)				
C	H	N	O[b]	S
47.61	6.27	0.55	43.89	0.23
Proximate analysis[a] (wt.%)				
Moisture content[c]	VM	fix-C	ash	HHV (MJ kg^{-1})
5.18	81.08	17.47	1.45	18.63

[a]Dry mass basis.
[b]By difference.
[c]As received;
fix-C: fixed carbon; HHV: higher heating value; VM: volatile matter.

Table 2. Specifications of instrument and crucible.

Instrument	Crucible number	Crucible volume (μL)	Crucible geometry (d × h*, mm)
TA SDT Q600	1	90	6 × 4

*d-crucible diameter; h- crucible depth.

The atmospheric pressure, thermogravimetric analyser TA SDT 600 was employed to perform thermogravimetric analysis (TGA) and differential thermal analysis of the corn cob powder. Table 2 summarises the specifications of TA SDT 600, the geometry and depth of the crucibles and the pans used with each TGA.

TA SDT 600 experiments employed nitrogen (N_2) (99.999% pure) as a purge gas with a flowrate of 100 mL min^{-1}. Prior to each experiment, a measured amount of corn cob powder (5 mg) was loaded into the appropriate open crucible. Each experiment started with a 30 min purging period at room temperature in order to bring out the evolved gases as soon as possible. Then, the reactor was heated to 105 °C and the temperature held for 30 min in order to remove the moisture, and finally raised up to the final temperature of around 950 °C at a heating rate of 10 °C min^{-1} and subsequently cooled to 50 °C at 100 °C min^{-1}. Each TGA experiment was normalised by the initial dry sample mass. For this purpose, the sample mass measured at 105 °C was selected as the initial dry sample mass.

Corn cob pyrolysis behaviour

Figure 1 presents the overall thermogravimetric decomposition process of corn cob powder. The derivative thermogravimetric (DTG) curves of corn cob pyrolysis contain shoulders and tailing, which indicated that more than one reaction was involved and that the corn cob consisted of different components (mainly of cellulose, hemicellulose and lignin) with different reactivity during pyrolysis. From the DTG data, the first loss of weight during the corn cob pyrolysis started at approximately 150 °C, owing to loss of bound moisture. The moisture evolution occurred at two distinct temperature regions. The bound water is present in the cell walls and it is believed to be hydrogen bonded to the hydroxyl groups of the main constituents of biomass: cellulose, hemicelluloses and, to a lesser extent, lignin (Siau, 2011). The DTG curves (Figure 1) show that there are clearly two areas of weight loss (between 250 °C and 400 °C) producing two peaks on the DTG curves. The first peak (around 300 °C) is owing to evolution of free water and owing to decomposition of hemicellulose; while the second peak (around 350 °C) is owing to decomposition of cellulose. Lignin decomposition occurs throughout the whole temperature range, but the main area of weight loss occurs at the higher temperatures, which means that lignin is mainly responsible for the tailing section, which was observed for all the biomass samples at the higher temperature. Clearly, a slight tail peak exists around 400 °C (Figure 1). The pyrolytic process was almost completed at approximately 550 °C, with charcoal yield at 550 °C being around 25 wt.%.

Empirical model development

The chemical formula of biomass feedstock is denoted as $CH_XO_YN_ZS_R$, and it can be calculated from the elemental analysis of the biomass and the mass fractions of C, H, O, N and S. In this model, the pyrolytic products, including the combined effect of both primary pyrolysis and secondary conversion, include the following components: charcoal, tar and gas. Also, some assumptions were necessarily made.

1. Ash is inert and it is not involved in any reaction.
2. Charcoal leaving pyrolysis units is considered to be composed of carbon, hydrogen and oxygen.
3. Gas products contain carbon-dioxide (CO_2), carbon-monoxide (CO), methane (CH_4), hydrogen (H_2) and water vapour (H_2O) and gas products are considered as ideal gas ($pV = nRT$).

These pyrolysis products depend on many parameters, such as biomass composition, biomass particle size, pyrolysis temperature, residence time, heating rate and pressure. However, based on literature research, pyrolysis temperature is the predominant factor responsible for variation of the yield and for the quality of

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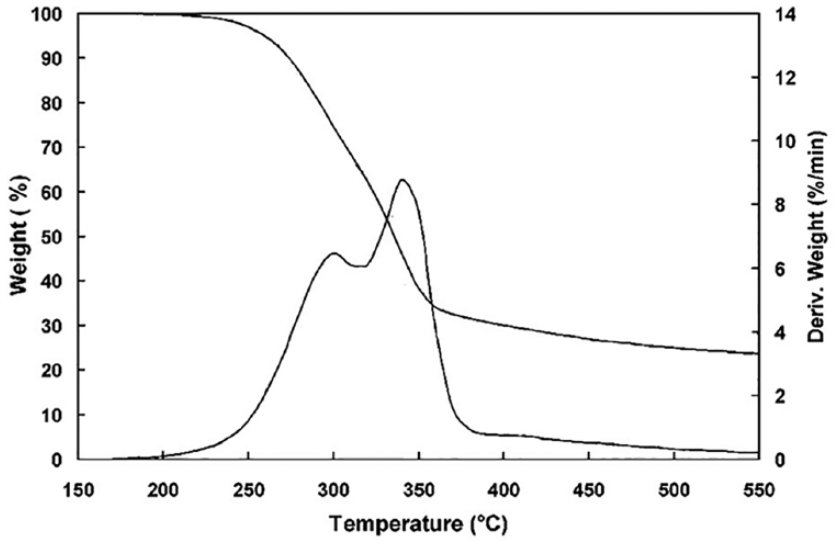


Figure 1. The weight loss and weight loss rate during corn cob pyrolysis.

the pyrolysis products (Brownsort, 2006; Di Blasi, 1996; Fakir, 2012; Horne and Williams, 1996; Ibrahim, 2012; Ioannidou et al., 2009; Kersten et al., 2005; Lee et al., 2005; Mullen et al., 2010b; Muylaert et al., 1999; Neves et al., 2011; Pütün et al., 1996; Pütüna et al., 1996; Renken, 2010; Scott et al., 1985; Senelwa, 1997; Şensöz and Angın, 2008; Smetsa et al., 2001; Srivastava et al., 1996; Sun et al., 2010; Tsaia et al., 2006).

In order to determine an empirical relationship between the product yield and pyrolysis temperature, data from studies covering most available agricultural residues (wheat straw, rice husk, grape husk, corn stover, corn cob and corn stalks), with particles having a variety of shapes and sizes (roughly between 0.2 and 250 mm) and reactor peak temperatures from 200 °C–1000 °C were analysed (Table 3). This analysis included different reactor types (tube, fixed bed, etc.), reactor scales (laboratory, pilot, industrial, etc.) and pyrolysis processes (slow, fast and flash) (Table 4).

Product distribution and characterisation

The recorded data on product distribution as a function of reactor peak temperature, including data for dry ash-free charcoal, tar and gas are presented in Figure 2.

Each sub-figure in Figure 2 consists of two data series; one for a slow heating rate and another for a fast heating rate. Our findings were similar to those of Neves et al. (2011), as the variability in data was high, with differences between lowest and highest yield as high as ±30 wt.% of parent biomass. The variability in yield is owing to differences in biomass (different biomass type, particle size, sample mass, etc.), reactor configuration and pyrolysis conditions (pressure, gas environment, gas flowrate, temperature programme, heating rate, residence time, etc.).

Antal and Mok (1990), and later Antal and Grønli (2003), compiled a succinct summary of the pyrolysis parameters and their influence on charcoal production. The most significant parameters in biomass pyrolysis were temperature, pressure, solid and volatile residence times, sample mass, particle size, biomass composition and heating conditions. Increasing the temperature and the solid residence time favours the formation of volatile and gaseous products (Di Blasi, 2008). As the particle size increases, the time necessary to achieve a certain conversion level at a certain temperature also increases. The volatile residence time can influence the process, as the volatile products can produce secondary interactions with the hot charcoal, which acts as a catalyst (Antal et al., 1996, 2000; Antal and Grønli, 2003; Di Blasi et al., 2008). Changes in heating conditions can modify the actual pathway and the rate of the reactions and affect the pyrolysis yields. Also, the biomass components react independently and, therefore, the thermal behaviour of biomass is also reflected by the individual behaviour of the biomass components. Each type of biomass has unique pyrolysis characteristics, by virtue of the specific proportions of the components present within it. Even if the components are the same chemical species, they can have differing reactivity if their pyrolysis is influenced by other intrinsic factors (chemical composition, ash content and composition, particle size and shape, density, moisture content, etc.) and/or by different process parameters.

Further, each TGA has different types of sample holders (different geometry, material, depth, etc.). The type of material and shape of the sample holder (crucible/pan) used for a measurement can have a large effect on the quality of the experimental results obtained. Deep crucibles can restrict gas flow more than flat ones. Also, reactions in the gas phase proceed more rapidly in completely open crucibles than in a so-called self-generated atmosphere. In a sealed crucible with a very small hole in the lid, or in a crucible with a lid without a hole placed loosely over the sample, the weight loss is shifted to a higher temperature (Gabbott, 2008).

Also, each data-point in Figure 2 is itself an average value taken from the results of a limited number of replicated experiments. According to Neves et al. (2011), pyrolysis experiments on identical materials and under identical conditions, in general,

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Table 3. Characteristics of agricultural residues (literature data and current research – corn cob experiment).

Raw material	Ultimate analysis (wt.%)						Proximate analysis (wt.%)				HHV	Reference
	C	H	N	O	S	P	Moisture	Volatile	Fix-C	Ash	MJ kg^{-1}	
Literature data for empirical model development												
Maize[a]	46.9	5.4	47.4	0.2	0.06			82.3	14.8	2.9	15.4	(Encinar et al., 1997)
Sunflower[a]	44.2	5.4	0.7	50.4	0.1			76.2	13.3	10.5	11.5	(Encinar et al., 1997)
Grape[a]	49.9	5.8	0.7	43.5	0.06			74	19.9	6.1	11.7	(Encinar et al., 1997)
Tobacco[a]	47.8	5.9	0.6	45.6	0.1			74.4	19.2	6.4	15.6	(Encinar et al., 1997)
Corn cob[a]	47.35	5.9	0.69	38.07	0.18					1.94	17.8[c]	(Mullen et al., 2010a)
Corn stover[a]	46.6	4.99	0.79	40.05	0.22					4.88	18.3[c]	(Mullen et al., 2010a)
Corn cob[a]	47.6	5	0	44.06				85.4		2.8	15.65	(Raveendran et al., 1995)
Wheat straw[a]	47.5	5.4	0.1	35.8				83.9		11.2	17.99	(Raveendran et al., 1995)
Rice straw[a]	36.9	5.0	0.4	37.9				80.2		19.8	16.78	(Raveendran et al., 1995)
Rice husk [a]	38.9	5.1	0.6	32.0				81.6		23.5	15.29	(Raveendran et al., 1995)
Corn stover[c]	50.1	5.01	0.93	33			9[d]			11		(Scott et al., 1985)
Wheat straw[c]	48.5	5.13	0.50		41.3		6.5[d]			4.6		(Scott et al., 1985)
Straw	42.69	6.04	0.46	47.11						3.7	17.53	(Fagbemi et al., 2001)
Grape residues	47.9	6.2	2.11		0.09		9[e]			5.1	23.83	(Di Blasi et al., 1999)
Rice husks	40.3	5.7	0.3		0.03		7[e]			15.3	18.73	(Di Blasi et al., 1999)
Wheat straw	43.6	6.2	0.3		0.08		7[e]			5.5	22.66	(Di Blasi et al., 1999)
Rapeseed[c]	62.1	9.1	3.9	24.9			4.9[e]	81.7[f]	7.9[f]	5.5[f]	26.7[f]	(Onay and Kockar, 2003)
Rice husk[a]								64.8	15.81	19.39		(Iwasakiet al., 2014)
Wheat straw[c]	46.5	6.3	0.9	46.3			7.1			3.2	18.51	(Zanzi et al., 2002)
Literature data for empirical model validation												
Corn cob[a]	47.57	6.27	0.55	43.89	0.23		5.18[d]	81.08	17.47	1.45		Current research
Corn cob[c]	43.77	6.23		50			7.57	84.37		8.06	18.25	(Ioannidou et al., 2009)
Corn stalk[c]	43.8	6.42		49.78			6.44	91.26		2.3	18.17	(Ioannidou et al., 2009)
Corn cob[c]	47.63	4.91	0.84	37.72	0.14	2.94	4.87[e]	80.66[c]		6.23[f]		(Cao et al., 2004)
Corn cob[a]	49	5.4	0.4	44.6			0	84.6	15.4	1		(Demirbas, 2004)

[a]Dry basis.
[b]Wet basis.
[c]Dry ash free.
[d]Moisture content on a wet mass basis.
[e]Moisture content on a dry basis.
[f]As received.
HHV: higher heating value.

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Table 4. Experimental details of pyrolysis processes used in literature studies and current research.

Sample size	Reactor type	Heating rate	Operation temperature	Gas and gas flow	Reference
Literature data for empirical model development					
1 mm	Laboratory TGA	Slow pyrolysis	350 °C–950 °C		(Wang et al., 2011)
1<d<1.6	Tube-stainless steel reactor		400 °C–700 °C	N_2, 200 $cm^3 min^{-1}$	(Encinar et al., 1997)
2 mm	Bubbling fluidised bed	Fast pyrolysis	500 °C		(Mullen et al., 2010a)
100–250 mm	Packed bed pyrolyser	50 °C min^{-1}	500 °C	N_2	(Mullen et al., 2010a)
105–250 µm	Bench-scale fluidised bed	Flash pyrolysis	400 °C–650 °C	N_2	(Scott et al., 1985)
10 cm	Quartz tube reactor		500 °C–1000 °C	He	(Fagbemi et al., 2001)
1–3 mm	Quartz tube reactor	Slow (conventional) pyrolysis, 25 °C min^{-1}	277 °C–927 °C	N_2	(Di Blasi et al., 1999)
0.2–1 mm	Quartz tube reactor	Slow (conventional) pyrolysis, 25 °C min^{-1}	277 °C–927 °C	N_2	(Di Blasi et al., 1999)
0.5–1 mm	Quartz tube reactor	Slow (conventional) pyrolysis, 25 °C min^{-1}	277 °C–927 °C	N_2	(Di Blasi et al., 1999)
0.224–1.8 mm	Fixed-bed tubular reactor and tubular transport reactor	Slow pyrolysis, 30 °C min^{-1}	400 °C–550 °C	N_2, 50,100, 200 or 400 $cm^3 min^{-1}$.	(Onay and Kockar, 2003)
0.224–1.8 mm	Fixed-bed tubular reactor and tubular transport reactor	Fast pyrolysis, 300 °C min^{-1}	400 °C–550 °C	N_2, 100 $cm^3 min^{-1}$	(Onay and Kockar, 2003)
0.224–1.8 mm	Fixed-bed tubular reactor and tubular transport reactor	Flash pyrolysis	400 °C–550 °C	N_2, 100 $cm^3 min^{-1}$	(Onay and Kockar, 2003)
	Fluidised bed reactor	Slow pyrolysis, 10 °C min^{-1}	300 °C–1200 °C	N_2	(Ioannidou et al., 2009)
	Fluidised bed reactor	Fast pyrolysis, $\lesssim$1000 °C s^{-1}	300 °C–1200 °C	N_2	(Ioannidou et al., 2009)
	Free fall reactor	Rapid pyrolysis, 500 °C s^{-1}	800 °C–1000 °C	N_2, 15 L min^{-1}	(Zanzi et al., 2002)
<1 mm	Captive sample reactor	Fast pyrolysis, 52 °C s^{-1}	360 °C–730 °C	He	(Ioannidou et al., 2009)
<1 mm	Captive sample reactor	Fast pyrolysis, 45 °C s^{-1}	380 °C–680 °C	He	(Ioannidou et al., 2009)
Literature data for empirical model validation					
	Laboratory TGA	Slow pyrolysis	350 °C–550 °C		Current research
<1 mm	Fixed bed (non-catalytic)	Slow pyrolysis	500 °C–700 °C	N_2, 100 $cm^3 min^{-1}$	(Ioannidou et al., 2009)
	Tube- stainless steel reactor	Slow pyrolysis, 30 °C min^{-1}	277 °C–927 °C	N_2, 80 mL min^{-1}.	(Cao et al., 2004)

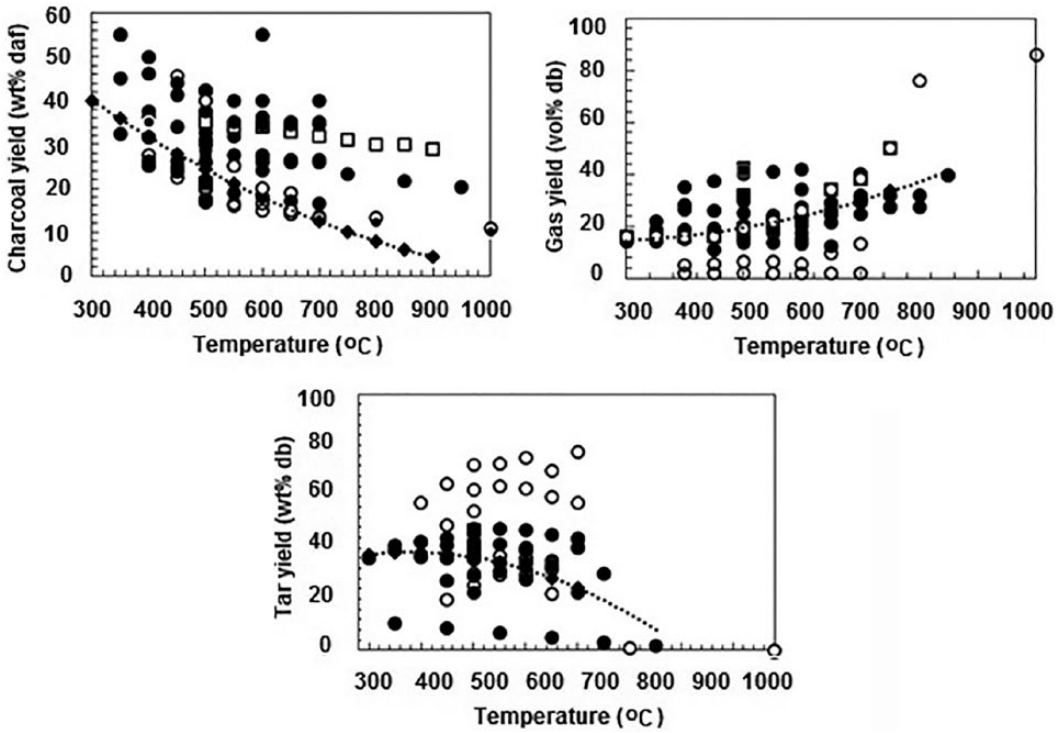


Figure 2. Yields of charcoal, gas and tar as a function of pyrolysis peak temperature, data points from literature (Cao et al., 2004; Demirbas, 2004; Di Blasi et al., 1999; Encinar et al., 1997; Fagbemi et al., 2001; Ioannidou et al., 2009; Iwasaki et al., 2014; Mullen et al., 2010a; Onay and Kockar, 2003; Raveendran et al., 1995; Scott et al., 1985; Zanzi et al., 2002).
Open circles: fast pyrolysis and small samples; black dots: slow pyrolysis and small samples; white square: fast pyrolysis and large samples; black square: slow pyrolysis and large samples.
Dashed line is given by equations (1)–(3).

do not yield identical results. Based on literature analysis, Neves et al. (2011) concluded that data on the yields of main products (charcoal, tar and gas) are often uncertain in roughly ±5%–20% of the reported values.

It has been established that high temperatures and rapid heating rates significantly lower charcoal yield when compared with slower heating rates (Figure 2(a)) (Hajaligol et al., 1982; Iwasaki et al., 2014; Zanzi et al., 2002). A study involving corn cob (Ioannidou et al., 2009) revealed that the total quantity of substrate that was decomposed increased (from 20 to 40 wt.%) when the heating rate was decreased. Long heating periods allow a sequence of secondary reactions to take place, whereas rapid heating tends to reduce these secondary reactions and the further degradation of the earlier formed pyrolysis products (Sadaka, 2011). Keowna et al. (2005) explained that recombination reactions occur to a lesser extent inside a particle under rapid heating conditions and, in the course of reactions between volatiles and charcoal, the 'self-gasification' of nascent charcoal by reactive components in the volatiles is favoured at rapid heating rates. Furthermore, Antal and Mok (1990) suggested that the reactor peak temperature has an unequivocal effect, not only on the yield, but also on the quality of the charcoal product. The charcoals produced at higher temperatures have a higher carbon content of both total- and fixed-carbon (Antal and Mok, 1990).

The yield of volatile products (gases and tars) increases with increasing temperature and heating rate, while the yield of solid residue decreases (Sadaka, 2011). Tar yields are higher with increased heating rate. Debdoub et al. (2006) observed that when the heating rate increased from 5 to 250 °C min^{-1} to 400 to 500 °C min^{-1}, the liquid yield increased from 45 to 68.5 wt.%. The maximum rate of devolatilisation increased almost linearly with increasing heating rate (Yu et al., 2007). Further, tar yields are higher with increased pyrolysis temperatures up to a maximum value, usually 450 °C to 600 °C, but this is dependent on equipment and other conditions (Bridgwater, 1999; Bridgwater and Cottam, 1992; Brownsort, 2006; Ibrahim, 2012; Zabaniotou et al., 2000). For fast pyrolysis, the peak liquid yields are generally obtained at a temperature of around 500 °C (Bridgwater, 1999). Peak liquid yields for slow pyrolysis are more variable. For example, Demirbas (2004) reported peak liquid yields of 28–41 wt.% at temperatures between 377 °C and 577 °C, depending on feedstock, when using a laboratory slow pyrolysis technique. Above these temperatures, secondary reactions causing depolymerisation of tar became more dominant, and tar yield decreased while gas yield increased. Also, the secondary decomposition of the charcoal at higher temperatures produces additional non-condensable gaseous product (Punsuwan and Tangsathitkulchai, 2014). Figure 2(b) shows the tar yield as a function of temperature.

Gas yields are generally low with irregular dependency on temperature when this is below the peak temperature for tar yield; above this, gas yields are increased strongly by higher

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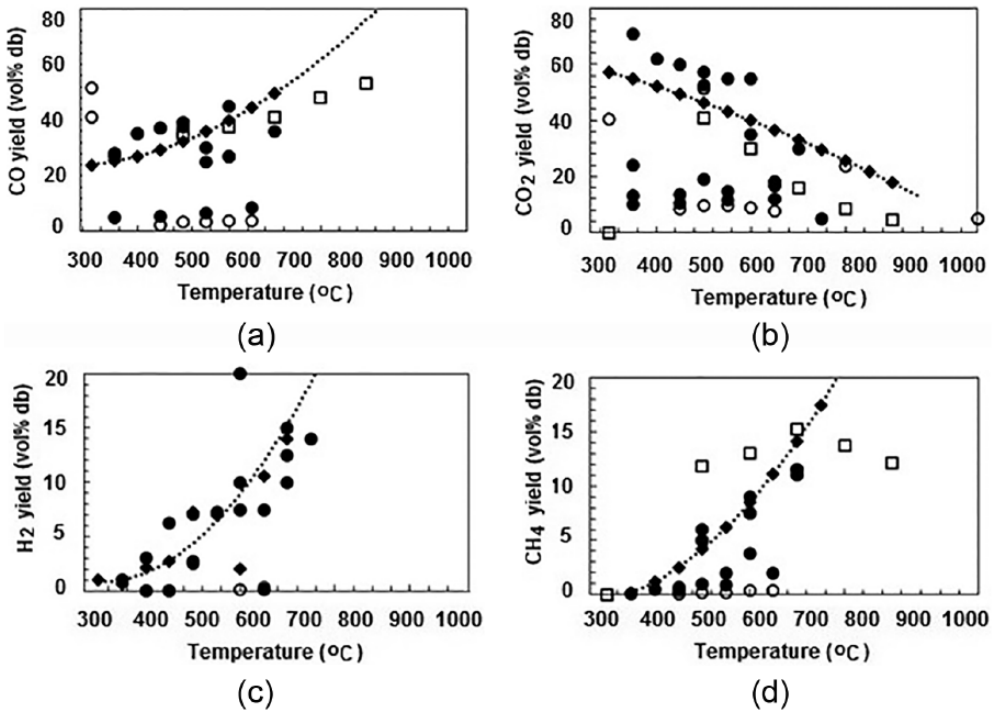


Figure 3. Gas yields for agricultural residues as functions of temperature (a) CO, (b) CO_2, (c) H_2, (d) CH_4 data points from literature (Cao et al., 2004; Demirbas, 2004; Di Blasi et al., 1999; Encinar et al., 1997; Fagbemi et al., 2001; Ioannidou et al., 2009; Iwasaki et al., 2014; Mullen et al., 2010a; Onay and Kockar, 2003; Raveendran et al., 1995; Scott et al., 1985; Zanzi et al., 2002). Open circles: fast pyrolysis and small samples; black dots: slow pyrolysis and small samples; white square: fast pyrolysis and large samples; black square: slow pyrolysis and large samples.
Dashed line is given by equations (4)–(7).

temperatures, as the main products of vapour decomposition are gases (Brownsort, 2006, 2009). The increase in gaseous products is predominantly owing to secondary cracking of the pyrolysis tar vapours at higher temperatures (Kim et al., 2013). At high pyrolysis temperatures, both the rate of primary pyrolysis and the rate of thermal cracking of tar to gaseous products are expectedly high. Furthermore, smaller particles produce higher gas yields because of the higher heat-up rate and heat flux as compared with larger particles. This observation agrees with the work of Wei et al. (2006), who studied the effect of particle size on product distribution from pyrolysis of pine sawdust and apricot stone in a free-fall reactor at 800 °C. The authors reported that the decrease of biomass particle size contributed to an increase in the gas yields. Figure 2(c) shows that the higher pyrolysis temperature led to more volatilisation, resulting in higher yield of gaseous products.

In order to obtain empirical correlations between pyrolysis process temperature and yield of pyrolysis products, literature data (listed in Table 3) were analysed using the weighted nonlinear least-squares curve-fitting method (dashed line in each subfigure in Figures 2 and 3).

The influence of pyrolysis temperature only on the yields of charcoal, tar and gas was taken into consideration because particle size and reactor type were found to be of limited influence on these empirical relationships, although this seems dependent on the biomass being pyrolysed. Temperature-dependent charcoal, tar and gas yields in the temperature range 300 °C–950 °C are described by empirical equations (1)–(3).

$$Ycc = 7.97T^2 \cdot 10^{-5} - 0.125 \cdot T + 68.87 \mathrm{R}^2 = 0.82 \quad (1)$$

$$Y_{TAR} = -1.38T^2 \cdot 10^{-4} + 0.12 \cdot T + 12.64 \mathrm{R}^2 = 0.89 \quad (2)$$

$$Y_{GAS} = 1.12T^2 \cdot 10^{-4} - 0.058 \cdot T + 30.77 \mathrm{R}^2 = 0.95 \quad (3)$$

where Ycc, Y_{TAR} and Y_{GAS} are the mass fraction of pyrolyisis products (charcoal, tar and gas) in dry ash free basis.

Properties of pyrolysis products

Gas composition. As mentioned above, an increase of the pyrolysis temperature consequently increases gas yield owing to the secondary reactions of pyrolysis vapours. According to Di Blasi et al. (1999), the pyrolysis gas consists mainly of CO_2 (the largest contribution), CO, CH_4, and lower amounts of H_2 and hydrocarbons. The yield of H_2, CO, CO_2 and CH_4 is plotted against temperature in Figure 3, which is a compilation of extracted literature data (Cao et al., 2004; Demirbas, 2004; Di Blasi et al., 1999; Encinar et al., 1997; Fagbemi et al., 2001; Ioannidou et al., 2009; Iwasaki et al., 2014; Mullen et al., 2010b; Onay and Kockar, 2003; Raveendran et al., 1995; Scott et al., 1985; Zanzi et al., 2002). Gas yield includes the activity of both the primary release of volatiles and secondary reactions.

Below 500 °C, when the activity of secondary reaction of volatile products is negligible, CO and CO_2 were the main gas species, with small quantities of CH_4 being produced. According to Di Blasi et al. (1999), this is attributable to the fact that CO_2 is a

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product of the primary pyrolysis of cellulose and hemicellulose by a pathway that becomes less favoured as the temperature increases. Previous studies (Fitzer, 1967; Yang et al., 2007) have shown that CO is formed during the primary decomposition of hemicellulose and cellulose, with a smaller proportion of CO coming from lignin. At low temperatures (below 500 °C), the production of H_2 is negligible for all types of biomass (Di Blasi et al., 1999). Production of other gases, H_2 and CH_4, increases with temperature increase. The high temperature favours the formation of H_2 at the expense of heavy hydrocarbons that are dehydrogenated from cracking (Di Blasi et al., 1999). Hydrogen is the main product of lignin pyrolysis.

Temperature-dependent yields of various gases in the temperature range 300 °C–950 °C are described by empirical equations (4)–(7). The influence of pyrolysis temperature only on the yields of gases, as in the case of charcoal and tar yields, was taken into consideration because particle size and reactor type were found to be of limited influence on these empirical relationships, although this seems dependent on the biomass being pyrolysed.

$$Y_{CO} = -2.65T^2 \cdot 10^{-4} + 0.27 \cdot T - 32.71 \quad R^2 = 0.64 \tag{4}$$

$$Y_{CO2} = -2.85T^2 \cdot 10^{-5} - 0.029 \cdot T + 70.89 \quad R^2 = 0.78 \tag{5}$$

$$Y_{CH4} = 6.69T^2 \cdot 10^{-5} - 0.037 \cdot T + 4.28 \quad R^2 = 0.98 \tag{6}$$

$$Y_{H2} = 7T^2 \cdot 10^{-5} - 0.0371T + 5.1117 \quad R^2 = 0.81 \tag{7}$$

where Y_{CO}, Y_{CO2}, Y_{CH4} and Y_{H2} are the volumetric fraction of different gases (CO, CO_2, CH_4 and H_2) in dry basis.

Charcoal composition. The elemental composition of charcoal varies roughly from that of the biomass itself to that of graphite (i.e. 100% carbon), being highly dependent on the pyrolysis conditions (Antal and Grønli, 2003; Antal et al., 1996, 2000; Neves et al., 2011). Charcoal becomes highly carbon rich with higher temperatures. The higher pyrolysis temperatures volatilise fewer carbon-containing compounds and act to fix carbon in the charcoal rather than volatilise it. The carbon content of the charcoal usually increases with temperature, being typically in the range of 85–95 wt.% (daf) above 800 °C (Antal and Grønli, 2003; Antal et al., 1996, 2000; Neves et al., 2011). According to Neves et al. (2011), the enrichment of carbon is accompanied by a loss of oxygen and hydrogen, amounts of which decrease to 5–15 wt.% and <2 wt.%, respectively. For corn cob, the maximum value of carbon content in charcoal is 87.71 wt.% at 950 °C (Brownsort, 2006; Wang et al., 2011). The temperature-dependent carbon, hydrogen and oxygen composition of charcoals ($Y_{j,ch}$) in the temperature range of 350 °C–950 °C is given by equations (8)–(10).

$$Y_{C,cc} = 7.77T^2 \cdot 10^{-5} - 0.08 \cdot T + 90.51 \quad R^2 = 0.83 \tag{8}$$

$$Y_{H,cc} = -1.72T^2 \cdot 10^{-6} - 0.002 \cdot T + 5.47 \quad R^2 = 0.87 \tag{9}$$

$$Y_{O,cc} = -9.11T^2 \cdot 10^{-5} + 0.10 \cdot T - 1.65 \quad R^2 = 0.85 \tag{10}$$

where $Y_{C,cc}$, $Y_{H,cc}$ and $Y_{Q,cc}$ are the mass fraction of chemical elements (carbon, hydrogen, oxygen, nitrogen) in charcoal in a dry ash free basis.

Tar composition. The data on the elemental composition of tar were compiled from different literature sources (Agblevor and Mante, 2011; Agblevor et al., 1995; Beis et al., 2002; Fagbemi et al., 2001; Mullen et al., 2010a, 2010b; Schröder, 2004).

It should be noted that it is difficult to analyse the elemental composition of tar, as some of the tar components volatilise and the mass of the tar sample, thus, does not remain constant throughout the course of the analysis (Schröder, 2004). Also, data are more abundant for the temperature range between 400 °C–600 °C, since investigations are usually focused on the characterisation of liquid products at operating conditions that maximise the yield of tar (Neves et al., 2011). Nevertheless, the composition of tar is influenced by the biomass type, temperature and the heating rate. The oxygen and carbon content of tar decrease slightly with increasing temperature, whereas the hydrogen content remains constant (Ateş et al., 2004; Beis et al., 2002; Neves et al., 2011; Nunn et al., 1985; Schröder, 2004; Weon, 2014). This indicates that the carbon, hydrogen and oxygen composition of tar is relatively close to that of the parent fuel, and is highly oxygenated. According to Neves et al. (2011), Schröder (2004) and Nunn et al. (1985), during primary pyrolysis, biomass undergoes low temperature decomposition into smaller tar molecules without significant change of the original chemical structure, while at higher temperatures, during secondary pyrolysis, tar becomes enriched in less oxygenated and likely more stable species.

In addition to these correlations, the energy, mass, and molar balances for each element (carbon, hydrogen, oxygen and nitrogen) were set and used to calculate pyrolysis products. The energy balance was formulated to include the overall heat loss of the pyrolysis unit. This estimation of the heat losses can be fixed by the user as a percentage of the product of dry biomass mass flow entering the system (kg h^{-1}) and its lower heating value (LHV) (kJ kg^{-1}).

Overall mass and elemental balance in the pyrolysis process

Antal (1985a) pointed out in his reviews of biomass pyrolysis theory and experimentation that a particle undergoing thermal decomposition must pass through certain temperature zones regardless of the final reaction temperature. Boundaries between the temperature stages are not sharp; there is always some overlap. During the initial phase of biomass heating at low temperature (≤200 °C), the free moisture and some loosely bound water is released. Also, some of the volatile products, such as acetic acid and formic acid, are released. At these low temperatures, only small fractions of non-condensable gases, such as CO and CO_2, are evolved (Antal, 1985a, 1985b; Karamarković, 2003; Sadaka, 2011). In the initial stage (200 °C–270 °C), biomass decomposition is more vigorous with the release of pyroligneous acids, water and non-condensable gases (CO, CO_2) (Basu, 2010;

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Sadaka, 2011). In the primary pyrolysis stage (270 °C–600 °C) (Basu, 2010), endothermic chemical decomposition of the main components of the biomass (cellulose, hemicelluloses, lignin and extractives) occurs. The biomass particles decompose into charcoal (primary charcoal), with release of combustible volatile products (CO, CH_4 and H_2) and formaldehyde, formic acid, methanol and acetic acid (Antal et al., 1985a, 1985b; Karamarković, 2003; Sadaka, 2011). During the final stage or secondary pyrolysis (600 °C–900 °C), if the reaction products are not removed from the reaction zone as soon as they are formed, secondary reactions begin and charcoal–gas reactions start (decomposition of volatiles into charcoal and non-condensable gases) (Fagbemi et al., 2001). If condensable gases are removed quickly from the reaction site, they condense outside in the downstream reactor as tar (Basu, 2010). All of these reactions are also severely affected by the catalytic effect exerted by the minerals present in the biomass and the charcoal.

The pyrolysis process can be described by:

$$\sum_i \boldsymbol{Y}_{i,biomass,af} + \mathbf{Y}_A \xrightarrow[\text{heat}]{} \sum_j \boldsymbol{Y}_{j,products,daf} + \boldsymbol{Y}_A + \boldsymbol{Y}_{H_2O} \qquad (11)$$

where $\sum_i Y_{i,biomass,af}$ is the mass fraction of the ith element (carbon, hydrogen, oxygen, nitrogen) in biomass in an ash free basis, and Y_A is ash content in biomass in a dry basis.

Pyrolysis products and gas composition can be described by:

$$\sum_j \boldsymbol{Y}_{j,products,daf} = \boldsymbol{Y}_{CC} + \boldsymbol{Y}_T + \boldsymbol{Y}_G \qquad (12)$$

$$\boldsymbol{Y}_G = \boldsymbol{Y}_{CO2} + \boldsymbol{Y}_{CO} + \boldsymbol{Y}_{CH4} + \boldsymbol{Y}_{H2} \qquad (13)$$

where $\sum_j Y_{j,products,\,daf}$ is the mass fraction of jth pyrolysis product (charcoal, tar and gas) in a dry ash free basis and Y_G is the mass fraction of total pyrolytic (permanent) gas.

In order to evaluate the elemental composition of the pyrolysis products, the overall elemental mass balances for the pyrolysis process have been taken into consideration. The four elemental balance equations for four chemical elements (carbon, hydrogen, oxygen and nitrogen) are described by the following equations.

Carbon balance:

$$\boldsymbol{Y}_{C,Bdaf} = \boldsymbol{Y}_{C,G} + \boldsymbol{Y}_{C,CC} + \boldsymbol{Y}_{C,T} \qquad (14)$$

where $Y_{C,Bdaf}$ is the mass fraction of carbon in biomass in a dry ash free basis and $Y_{C,G}$, $Y_{C,CC}$ and $Y_{C,T}$ are the mass fraction of carbon in gas, charcoal and tar, respectively.

Hydrogen balance:

$$\boldsymbol{Y}_{H,Bdaf} = \boldsymbol{Y}_{H,G} + \boldsymbol{Y}_{H,CC} + \boldsymbol{Y}_{H,T} \qquad (15)$$

where $Y_{H,Bdaf}$ is mass fraction of hydrogen in biomass in a dry ash free basis and $Y_{H,G}$, $Y_{H,CC}$ and $Y_{H,T}$ are a mass fraction of hydrogen in gas, charcoal and tar, respectively.

Oxygen balance:

$$\boldsymbol{Y}_{O,Bdaf} = \boldsymbol{Y}_{O,G} + \boldsymbol{Y}_{O,CC} + \boldsymbol{Y}_{O,T} \qquad (16)$$

where $Y_{o,bdaf}$ is mass fraction of oxygen in biomass in a dry ash free basis and $Y_{O,G}$, $Y_{O,CC}$ and $Y_{O,T}$ are a mass fraction of oxygen in gas, charcoal and tar, respectively.

Nitrogen balance:

$$\boldsymbol{Y}_{N,Bdaf} = \boldsymbol{Y}_{N,G} \qquad (17)$$

where $Y_{N,Bdaf}$ is a mass fraction of nitrogen in biomass in a dry ash free basis and $Y_{N,T}$ is a mass fraction of nitrogen in gas.

Overall energy balance in the pyrolysis process

The overall energy balance for the pyrolysis of 1 kg of biomass was defined as:

$$\begin{aligned} &\boldsymbol{Y}_{Bdaf\cdot} \cdot \boldsymbol{LHV}_{bio} \\ &= \boldsymbol{Y}_{CC\cdot} \cdot \boldsymbol{LHV}_{CC} + \boldsymbol{Y}_{T\cdot} \cdot \boldsymbol{LHV}_T \\ &+ \boldsymbol{Y}_{G\cdot} \cdot \left(\boldsymbol{LHV}_G + \sum_j \boldsymbol{x}_i \int_{To}^{Tp} \boldsymbol{c}_{p,i}\, \boldsymbol{dT} \right) \\ &+ \boldsymbol{Y}_{H_2o\cdot} \left(\boldsymbol{h}_{H_2O} + \int_{To}^{Tp} \boldsymbol{c}_{p,H_2O} \boldsymbol{dT} \right) \end{aligned} \qquad (18)$$

where $Y_{Bdaf\cdot}$ is the mass of biomass in a dry ash free basis, $Y_{H_2o\cdot}$ is the mass fraction of water vapour, h_{H_2O} is the enthalpy of water vapour, c_{p,H_2O} is the specific heat capacity of water vapour, $c_{p,i}$ is the specific heat capacities of the ideal gases (H_2, CO, H_2O, CO_2, CH_4) at constant pressure and LHV_{bio}, LHV_{CC}, LHV_T and LHV_G are the lower heating value of biomass, charcoal, tar and gas, respectively. Also, To and Tp are the start and ending pyrolysis temperature.

The specific heat capacities of the ideal gases at constant pressure were calculated by the third-order polynomial equations taken from literature and presented in Table 5 (Balmer, 2011; Karamarkovic and Karamarkovic, 2010).

Modelling software

Calculation of pyrolysis products, components and energy values, using the system of equations above, was conducted using the engineering equation solver (EES). The EES software has been found to be very suitable for modelling this kind of system, since it contains all necessary thermodynamic functions (enthalpy, molar mass and density, etc.), which can make the model user-friendly (Puig-Arnava et al., 2012). To apply the model, the following input data were taken: the elemental composition of the biomass and the reactor temperature.

Validation of the model

In order to validate the model, a comparison of model predictions of temperature effects on the distribution of pyrolysis products in the current study and previous studies (Cao et al., 2004; Demirbas, 2004; Ioannidou et al., 2009) was summarised. This procedure

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Table 5. Specific heat capacities of ideal gases at constant pressure (Balmer, 2011).

Gas	c_p –KJ / Kmol,θ–(T+273) / 100	Range °C	Max. error %
H_2	$c_p = 56.505 - 702.74\theta^{-1} + 1165.0\theta^{-1} - 560.70\theta^{-1.5}$	27–3227	0.60
CO	$c_p = 69.145 - 0.70463\theta^{0.75} - 200.77\theta^{-0.5} + 176.76\theta^{-0.75}$	27–3227	0.42
H_2O	$c_p = 143.05 - 183.54\theta^{0.25} + 82.751\theta^{0.5} - 3.6989\theta$	27–3227	0.43
CO_2	$c_p = -3.7357 + 30.529\theta^{0.5} - 4.1034\theta + 0.0241980\theta^{2}$	27–3227	0.19
CH_4	$c_p = -672.87 + 439.74\theta^{0.25} - 24.875\theta^{0.75} + 323.88\theta^{-0.5}$	27–1727	0.15

Table 6. Comparison of present study methods and results with literature data.

	Current research		Reference					
	Model	Experiment	(Ioannidou et al., 2009)		(Ioannidou et al., 2009)		(Cao et al., 2004)	(Demirbas et al., 2004)
Reactor type			Captive sample		Fixed bed (non-catalytic)			
Biomass	Corn cob		Corn cob	Stover	Corn cob	Stover	Corn cob	Corn cob
Operation temperature (°C)	350–550		360–730	380–680	500–700		600	677–977
Product yields (wt.% daf)								
Gases	23.96–32.28		14–63	22–55	16–40	14–29	27–41	
Liquid	39.67–45.17		23–15	20–11	22–40	29–42	42–35.5	
Charcoal	36.37–22.45	38.7–22.5	48–17.6	49–22	37–55	33–35	32–25	31.8–19.1
Water and losses								36.7–6
Gas composition (v/v %)								
CO	32.94–40.36		41–51	35–59	37–44	27–37	27–30	
CO_2	59.62–30.22		24–3.5	39–2	31–52	30–52	71–52	
H_2	0.5–8.85		28–42.5	18–28	2–13	2–15	1–7	
CH_4	3.58–14.81		7–9	8–12	4–13	5–11	0–3	
LHV of gas (MJ m^{-3})	10–13		13–15					

LHV: lower heating value.

was adopted because of the limited experimental data for simulated corn cob residues (i.e. gas and tar yield, gas composition, etc.). The error in this comparison is estimated by the root mean squared error (RMSE) defined as in:

$$\boldsymbol{RMSE} = \sqrt{\frac{\sum_{i=1}^{n}\left(\boldsymbol{X}_{obs,i} - \boldsymbol{X}_{model,i}\right)^2}{\boldsymbol{n}}} \qquad (19)$$

where $X_{obs,i}$ are observed values at place i, $X_{model,i}$ are modelled values at place i and n is the number of data.

To predict the results, the model parameters were adjusted by minimising the sum of the differences between the experimental and predicted results for gas and tar composition. At first, validation of the model was conducted by comparing predicted charcoal yield with experimental results from pyrolysis of corn cob (Figure 2). The model predicted measured yields within ±10% accuracy.

A comparison of predicted (modelled) and measured yields (experimental and literature data) of all pyrolytic products is provided in Table 6 and Figures 4 and 5, which focus on the yields of total charcoal, gas and tar produced by pyrolysis of corn cobs and stover. The model predicted most of the measured yields within ±25% accuracy, and the prediction was often within the uncertainty of the measurements. In general, the model overestimated the measured data, with the predicted yields of tar significantly higher than the observed yields.

The comparison between charcoal yields obtained from the literature, and experimental and predicted charcoal yields (the current study) is shown in Figure 5(a). From this figure, it can be concluded that the model predicts the charcoal yield with good accuracy, although the differences from values obtained in one study (Ioannidou, et al., 2009) were higher than for other studies. The RMSE values obtained were 2.14, 22.99, 2.97 and 1.66, with input values from Cao et al. (2004), Ioannidou et al. (2009), Demirbas (2004) and the current study experiments, respectively. The comparison between literature and experimental gas yields and predicted gas yields is shown in Figure 5(b). Generally, the model predicted the gas yields with good accuracy, although the differences from values obtained in Demirbas (2004) were higher than for other studies. The RMSE values obtained were 8.48, 13.74 and 10.9, for input values from Cao et al. (2004), Ioannidou et al. (2009) and Demirbas (2004), respectively. The proposed model significantly overestimated the tar yield measured by Demirbas (2004) (Figure 5(c)). For the other two studies, the model predicted the tar yields with good accuracy (±25% accuracy). The RMSE values obtained were 1.10, 9.09 and 28.31, for input values from Cao et al. (2004), Ioannidou et al. (2009) and Demirbas (2004), respectively. Figure 6 shows the composition of the dry gas product predicted by the model along with data extracted from the literature. The model predicted most of measured yields within ±25% accuracy, and the predictions were often within

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the uncertainty of the measurements. In general, the model underpredicted the measured data, with the predicted yields of CO_2 lower than the observed ones.

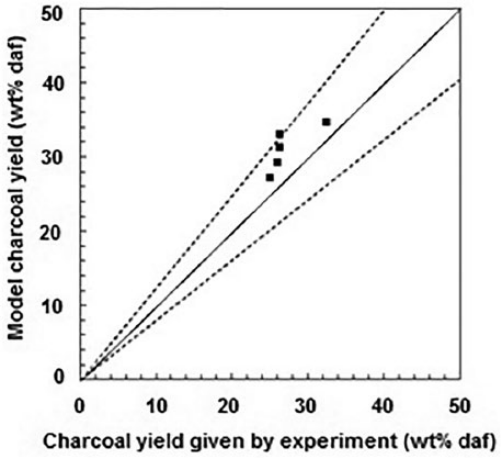


Figure 4. Modelled yield of charcoal compared with experimental yield of charcoal (corn cob pyrolysis).
Long dash line: 10% deviation.

Summary and conclusions

In this study, investigations on biomass pyrolysis under inert atmosphere, including experiments utilising several agricultural biomass residues in a variety of reactors and over a wide range of temperatures (300 °C–1000 °C) were analysed. Experimental data on pyrolytic product distribution and properties were produced in this study and were extracted from the literature and corn cob pyrolysis experiment, together with information concerning the experimental procedures and biomass properties. The pyrolytic products were dry, ash-free charcoal, tar and gas (H_2, H_2O, CO, CO_2, CH_4 and other light hydrocarbons).

The analysis of general trends of pyrolysis product distribution showed that temperature and heating rate are important process parameters in determining the yields of charcoal, tar and gas. At the low temperature range (⩽600 °C), biomass decomposes into primary charcoal with release of combustible gases (CO, CH_4 and H_2) and tar (Antal, 1985a, 1985b; Karamarković, 2003; Sadaka, 2011). With temperature increase, charcoal yield decrease is compensated by the production of tar. At the high temperature range (⩾600 °C), secondary pyrolysis reactions

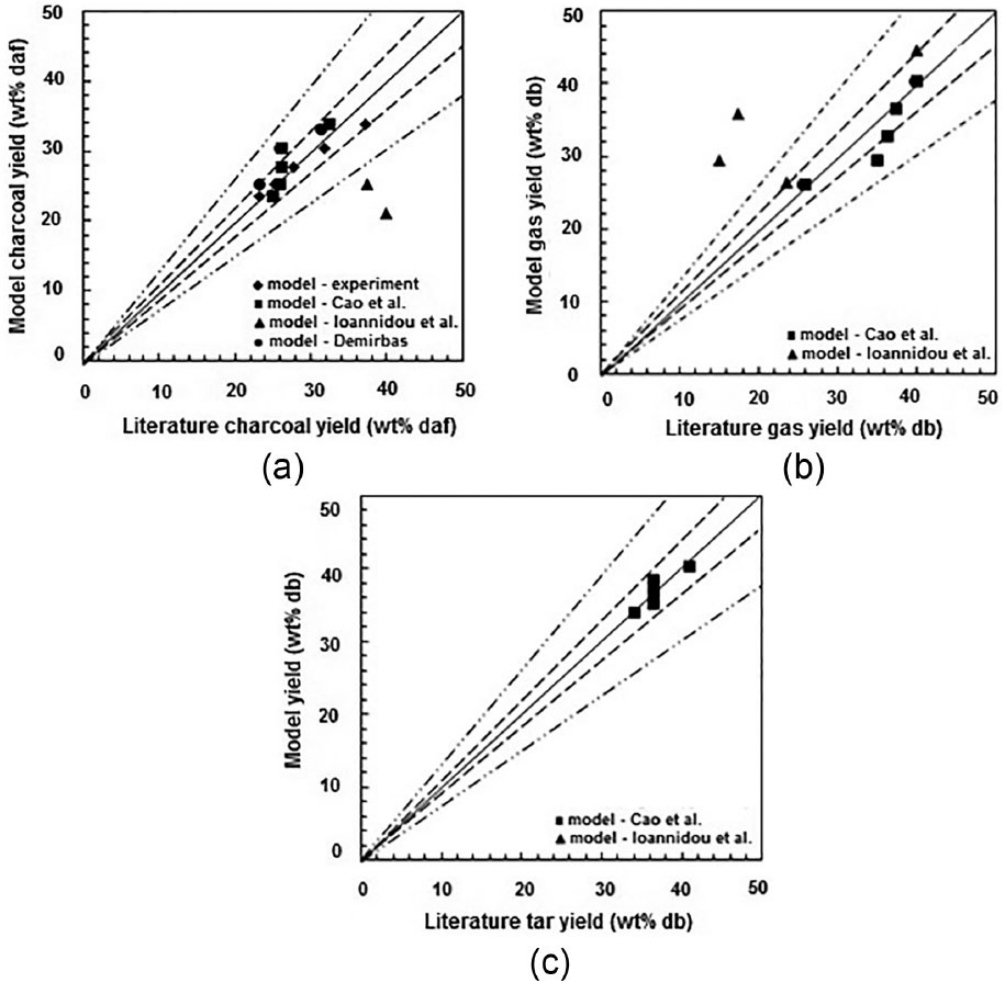


Figure 5. Modelled yield of charcoal, gas and tar compared with yields obtained from literature. (a) charcoal yield, (b) gas yield and (c) tar yield.
Long dash line: 10% deviation; long dash dot dot: 25% deviation.

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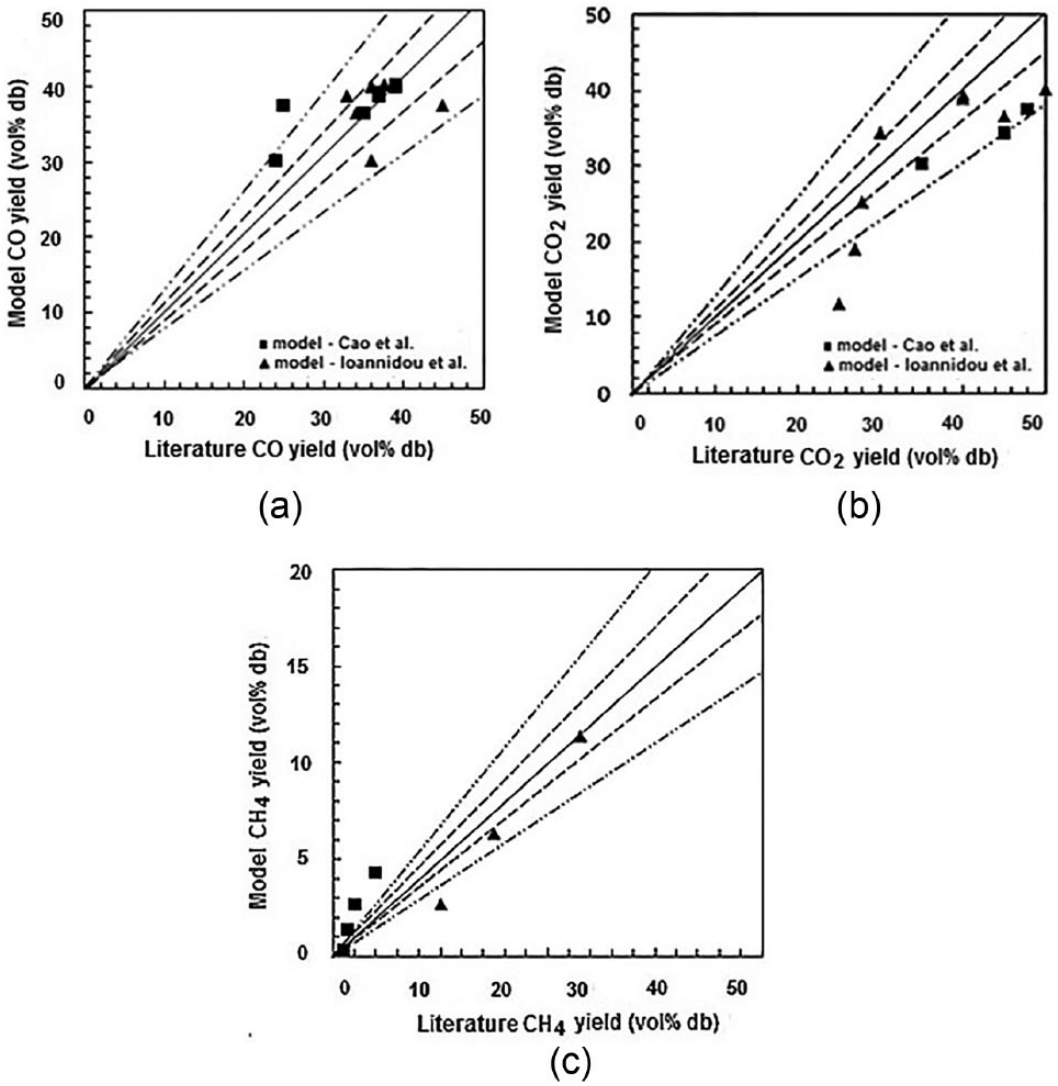


Figure 6. Modelled gas composition compared with gas composition obtained from experimental literature a) CO, b) CO_2, c) CH_4.
Long dash line: 10% deviation; long dash dot dot: 25% deviation.

occur. The yield of tar decreases with temperature increase, which is compensated by an increase in the yield of gases. Tars are converted into a variety of gaseous species. The yields of CO, H_2 and CH_4 show a similar pattern of change with temperature. The yield of CO_2 decreases slowly as temperature increases. Variations in the yield of charcoal with temperature changes are comparatively small.

The general pyrolytic product distribution trends were correlated with temperature using a novel empirical model. The empirical model of pyrolysis consisted of a system of equations in which the mass balance for each element and the energy balance of the pyrolysis process were combined with empirical parameters. The empirical equations used in the model were derived from the extracted literature data, and included relations between the temperature and pyrolytic products and composition yields. Comparison between predictions made by the developed model and measured data on the product distribution from pyrolysis of agricultural residues showed good agreement (within ±25% accuracy). In general, the model overestimated the measured data, with the predicted yields of tar significantly higher than the observed ones. Although this semi-empirical model needs further refinements, it would help to predict the behaviour of different biomass types, and would be a useful tool for preliminary calculations, design and operation of biomass pyrolysers. It would be also a useful first step in analysis of other systems. For example, it could be used as an input in the gasification and combustion model of an internal combustion engine or another gas to energy engine, or to model a whole biomass co- or tri-generation plant. Further refinements of the model would include coupling this novel semi-empirical model with kinetics models, and with transport equations, and considering intra- and extra-particle phenomena (e.g. shrinkage) and the operating conditions prevailing in practical pyrolysis reactors. The development of a model for the entire process of biomass pyrolysis would facilitate the analysis of conditions to optimise the pyrolysis process and to minimise its current drawbacks.

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